

Option D – Energy and the Environment

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)	Use of $\lambda_{\text{max}} = \frac{W}{T}$ (1) $\lambda_{\text{max}} = \frac{2.9 \times 10^{-3}}{288} = 1 \times 10^{-5} \text{ [m]}$ (1)	1	1		2	1	
		(ii)	Gases allow shorter λ solar radiation to pass [where it is absorbed by Earth] (1) Gases absorb [some of the] longer λ / infra-red / $10 \mu\text{m}$ radiation [emitted by Earth] (1) Gases emit radiation {in all directions / back to Earth} (1)	3			3		
	(b)	(i)	$\text{mass}_{\text{ice}} = 920 \times 3.0 \times 10^{11} = 2.76 \times 10^{14}$ and $V_{\text{water}} = \frac{2.76 \times 10^{14}}{1000} = 2.76 \times 10^{11}$ (1) $2.76 \times 10^{11} \times 20 = 5.52 \times 10^{12} \text{ [m}^3\text{]}$ (1) Alternatively for 1st marking point: $V_{\text{water}} = \frac{3.0 \times 10^{11}}{\frac{1000}{920}} = 2.76 \times 10^{11}$		2		2	2	
		(ii)	Darker areas will absorb more radiation / heat [than lighter areas] (1) Resulting in an increased temperature and rate of ice melting (1)		2		2		
	(c)	(i)	Hydroelectric relatively constant whereas PV increasing (1) No new hydroelectric sites available or commercially viable / advances in efficiency of PV cells / availability of [domestic] PV cells (1)			2	2		

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		(ii)	Use of $P_{\max} = IV$ [at / close to 'knee'] (1) $P_{\max} = 130 \pm 5$ [W] (1) $P_{\text{in}} = IA = 600 \times 1.2 = 720$ [W] (1) $\frac{130 \text{ ecf}}{720} = 0.18$ or 18 [%] (1)	1	1 1 1		4	3	
	(d)	(i)	Use of $\frac{\Delta Q}{\Delta t} = KA \frac{\Delta \theta}{\Delta x}$ (1) Answer seen 4900 (1)	1	1		2	1	
		(ii)	Attempt at calculating $\Delta \theta_{\text{for 1 layer}}$ (1) $\Delta \theta_{\text{carpet}} = \frac{1225 \text{ ecf} \times 10 \times 10^{-3}}{0.045 \times 60} = 4.5$ and $\Delta \theta_{\text{underlay}} = 5.8$ and $\Delta \theta_{\text{concrete}} = 3.5$ (1) Add to give 13.8~14 so claim is reasonable (1) Also accept use of $\frac{5\,000}{4} = 1250$ W Alternative: Attempt at calculating $\frac{\Delta x}{KA}$ for 1 layer (1) $R_{\text{T(carpet)}} = \frac{\Delta x}{KA} = \frac{10 \times 10^{-3}}{0.045 \times 60} = 0.0037$ and $R_{\text{T(underlay)}} = 0.0048$ and $R_{\text{T(concrete)}} = 0.0029$ (1) Add to give 0.0114 and $\frac{\Delta Q}{\Delta t} = \frac{\Delta \theta}{\frac{\Delta x}{KA}} = 1228 \approx 1225$ so claim is reasonable (1)			3	3	3	

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			Alternative: Attempt at using R values (1) $R_{\text{concrete}} = \frac{0.12}{0.7} = 0.17$ $R_{\text{underlay}} = \frac{0.01}{0.035} = 0.29$ $R_{\text{carpet}} = \frac{0.01}{0.045} = 0.22$ Total R value = $0.68 \text{ [K m}^2 \text{ W}^{-1}\text{]} (1)$ So $R_T = \frac{\frac{\Delta T}{\phi}}{A} \rightarrow \phi = \frac{\Delta TA}{R_T}$ $\phi = \frac{14 \times 60}{0.68} = 1235 \text{ [W]} \text{ so claim is reasonable (1)}$						
			Question 10 total	6	9	5	20	10	0